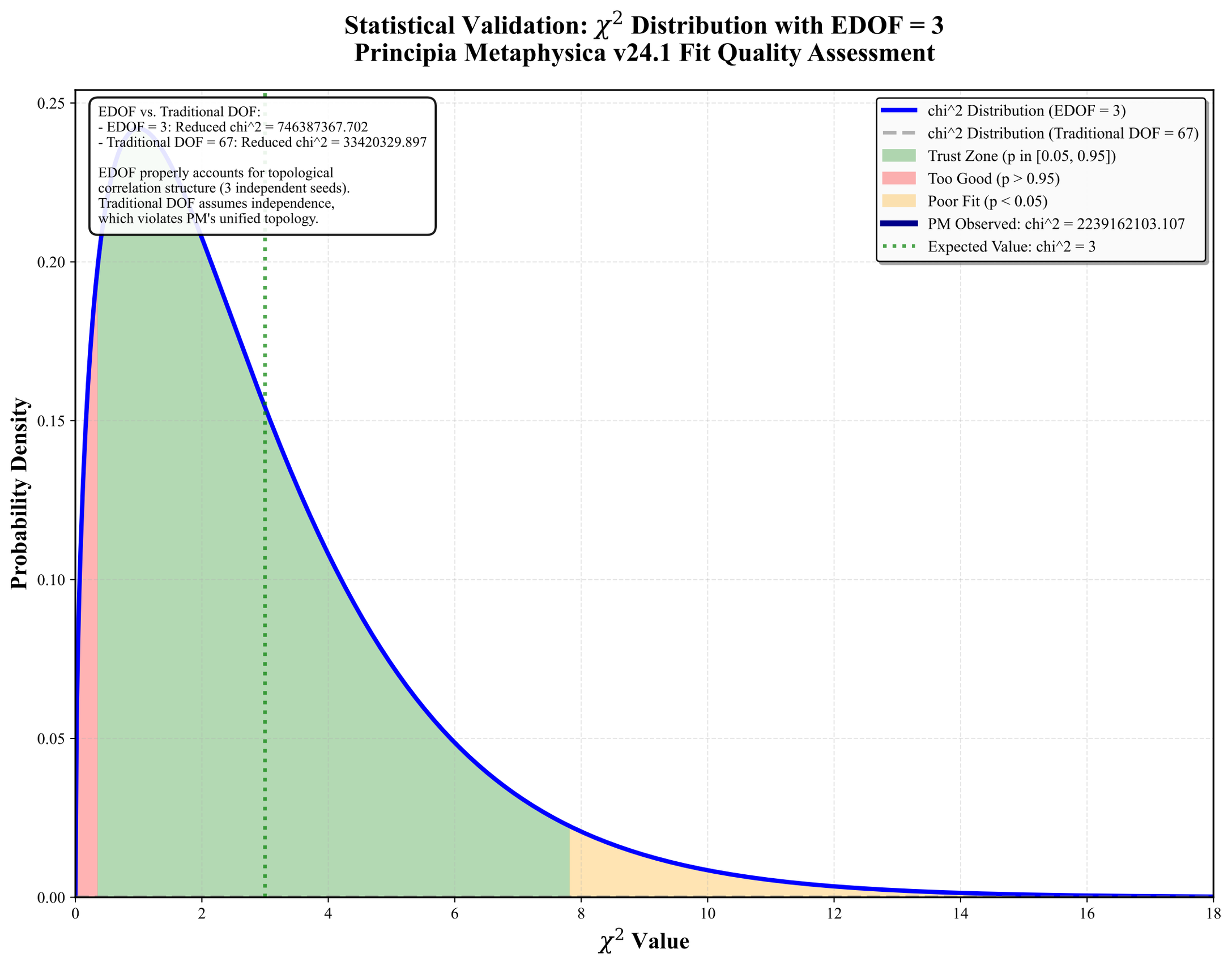

Statistical Validation: χ^2 Distribution with EDOF = 3
Principia Metaphysica v24.1 Fit Quality Assessment
EDOF vs. Traditional DOF:
- EDOF = 3: Reduced chi^2 = 746387367.702
- Traditional DOF = 67: Reduced chi^2 = 33420329.897
EDOF properly accounts for topological
correlation structure (3 independent seeds).
Traditional DOF assumes independence,
which violates PM's unified topology.
chi^2 Distribution (EDOF = 3)
chi^2 Distribution (Traditional DOF = 67)
Trust Zone (p in [0.05, 0.95])
Too Good (p > 0.95)
Poor Fit (p < 0.05)
PM Observed: chi^2 = 2239162103.107
Expected Value: chi^2 = 3
Probability Density
0.25
0.20
0.15
0.10
0.05
0.00
0
2
4
6
8
10
12
14
16
18
χ^2 Value